



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°9_ – SYSTEME

COURS: SYSTEME

PRÉNOM: Lens Sandro

NOM: PÉTIOTE

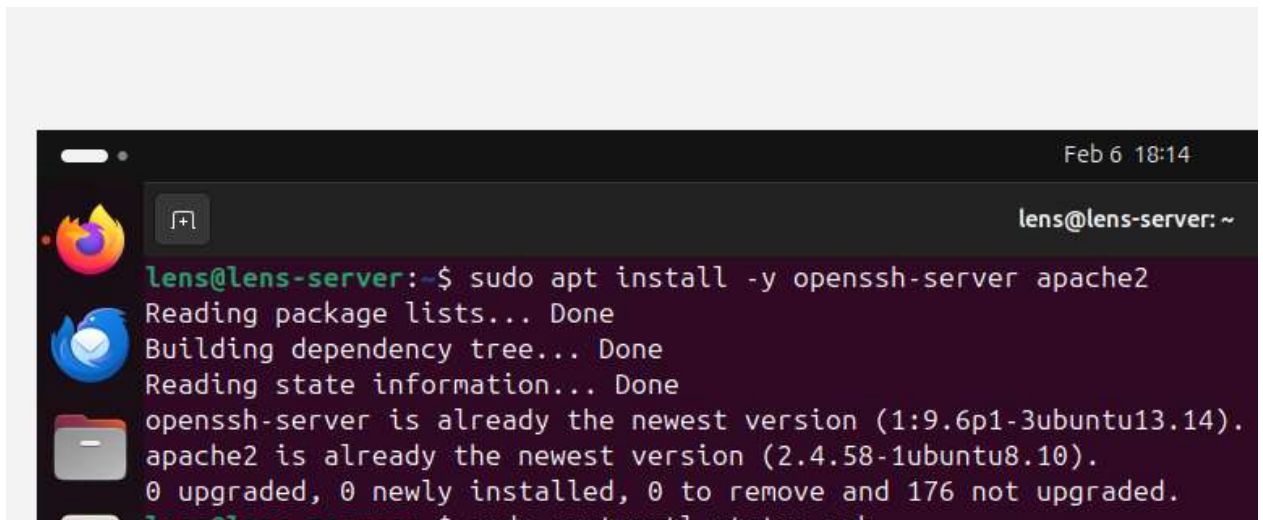
NIVEAU: L3

DATE: 06/02/2026

L'objectif de ce TD est de:

1. Comprendre ce qu'est un service / démon sous Linux
2. Savoir lister, démarrer, arrêter, activer et désactiver des services
3. Apprendre à surveiller et dépanner les services
4. Automatiser la gestion des services via scripts
5. Comprendre les principes de sécurité Linux
6. Mettre en place des comptes sécurisés et permissions correctes
7. Configurer le pare-feu et services essentiels
8. Analyser les logs et détecter les anomalies
9. Comprendre les ressources critiques du système : CPU, RAM, disque, réseau
10. Utiliser les outils Linux pour mesurer la charge système
11. Automatiser la collecte et la surveillance avec scripts Bash

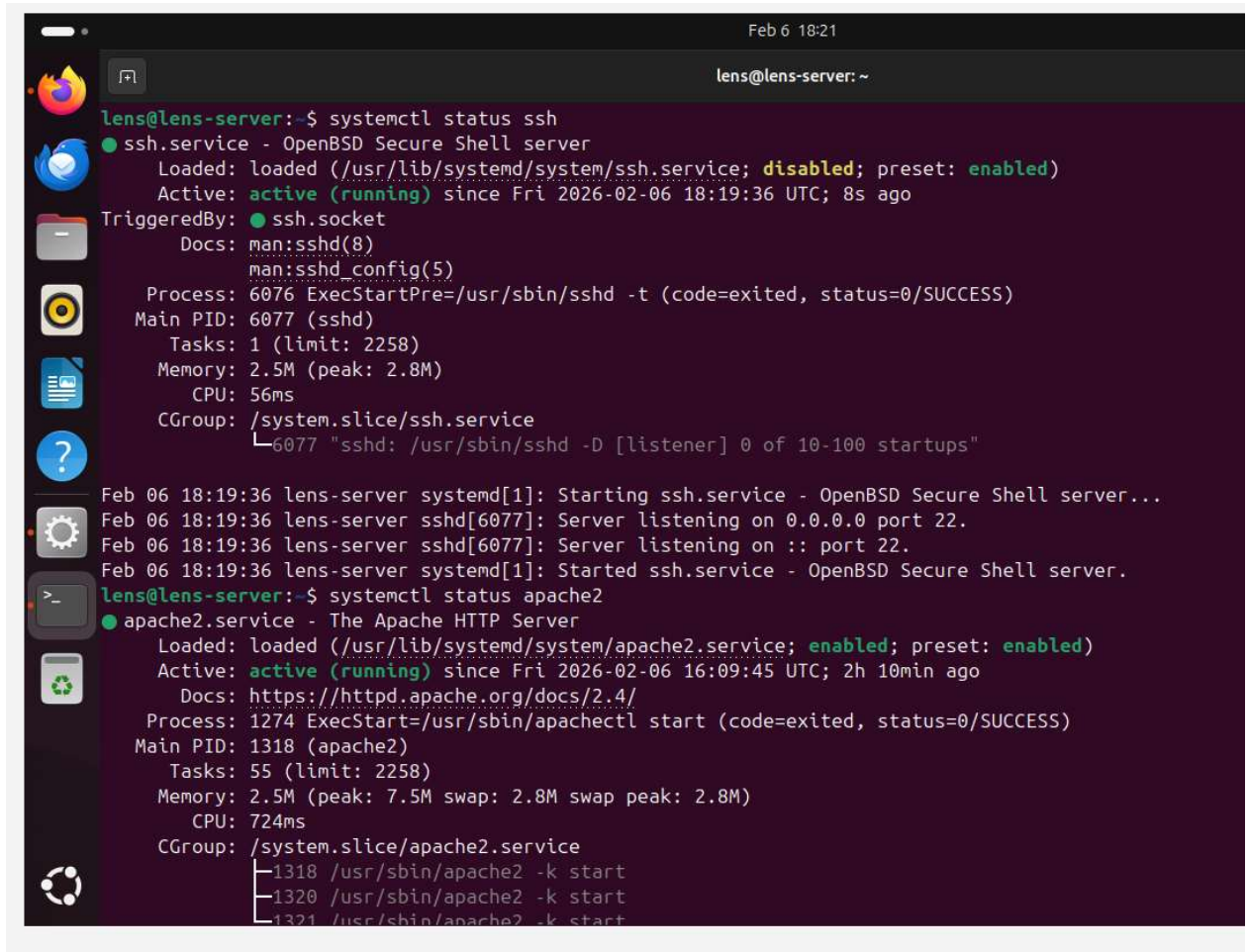
1. Installer et configurer **SSH** et **Apache**

A screenshot of a terminal window on a Linux system. The window title is 'Feb 6 18:14' and the prompt is 'lens@lens-server: ~'. The user has entered the command 'sudo apt install -y openssh-server apache2'. The output shows that both packages are already installed at the latest versions: 'openssh-server is already the newest version (1:9.6p1-3ubuntu13.14)' and 'apache2 is already the newest version (2.4.58-1ubuntu8.10)'. The summary line indicates '0 upgraded, 0 newly installed, 0 to remove and 176 not upgraded'.

```
lens@lens-server:~$ sudo apt install -y openssh-server apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssh-server is already the newest version (1:9.6p1-3ubuntu13.14).
apache2 is already the newest version (2.4.58-1ubuntu8.10).
0 upgraded, 0 newly installed, 0 to remove and 176 not upgraded.
```

Déjà installés et actifs

2. Vérifier l'état des services



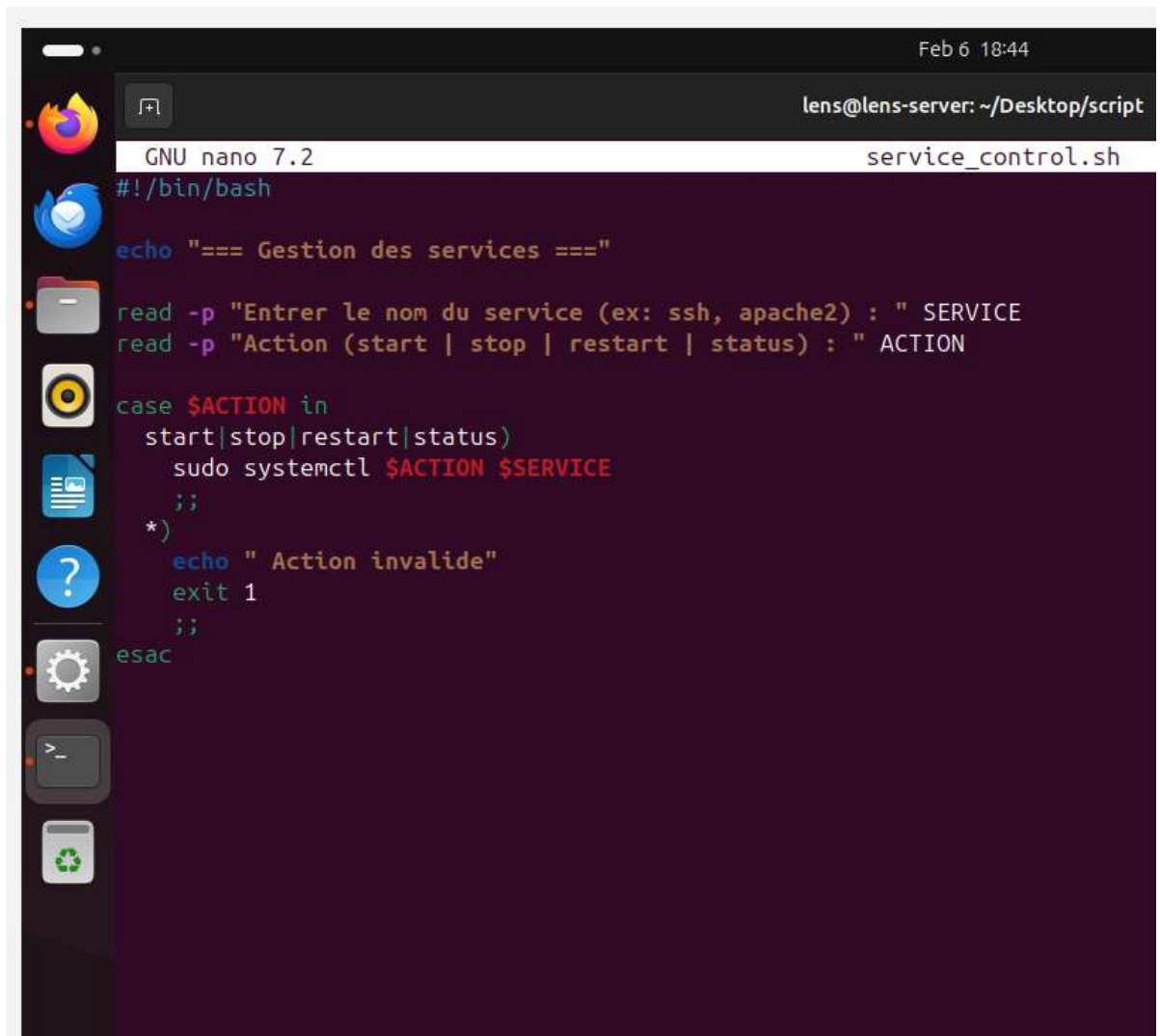
```
lens@lens-server:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Fri 2026-02-06 18:19:36 UTC; 8s ago
     TriggeredBy: ● ssh.socket
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 6076 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 6077 (sshd)
      Tasks: 1 (limit: 2258)
     Memory: 2.5M (peak: 2.8M)
        CPU: 56ms
    CGroup: /system.slice/ssh.service
            └─6077 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Feb 06 18:19:36 lens-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 06 18:19:36 lens-server sshd[6077]: Server listening on 0.0.0.0 port 22.
Feb 06 18:19:36 lens-server sshd[6077]: Server listening on :: port 22.
Feb 06 18:19:36 lens-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
lens@lens-server:~$ systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-02-06 16:09:45 UTC; 2h 10min ago
     Docs: https://httpd.apache.org/docs/2.4/
    Process: 1274 ExecStart=/usr/sbin/apachectl start (code=exited, status=0/SUCCESS)
   Main PID: 1318 (apache2)
      Tasks: 55 (limit: 2258)
     Memory: 2.5M (peak: 7.5M swap: 2.8M swap peak: 2.8M)
        CPU: 724ms
    CGroup: /system.slice/apache2.service
            └─1318 /usr/sbin/apache2 -k start
              └─1320 /usr/sbin/apache2 -k start
                └─1321 /usr/sbin/apache2 -k start
```

3. Créer un script pour démarrer ou arrêter un service

Création de mon fichier service_control.sh pour mettre le script et test du script pour la gestion d'un service

Mon script dans nano



The image shows a terminal window with a dark background. On the left is a vertical sidebar with icons for various applications: Firefox, a file manager, a terminal, a help icon, a settings icon, a terminal icon, and a trash icon. The terminal window title bar shows the date and time 'Feb 6 18:44' and the user and file path 'lens@lens-server: ~/Desktop/script'. The nano editor is open, editing a file named 'service_control.sh'. The editor's status bar at the top indicates 'GNU nano 7.2'. The script content is as follows:

```
#!/bin/bash

echo "=== Gestion des services ==="

read -p "Entrer le nom du service (ex: ssh, apache2) : " SERVICE
read -p "Action (start | stop | restart | status) : " ACTION

case $ACTION in
    start|stop|restart|status)
        sudo systemctl $ACTION $SERVICE
        ;;
    *)
        echo " Action invalide"
        exit 1
        ;;
esac
```

Test

```
Feb 6 18:43
lens@lens-server: ~/Desktop/script

lens@lens-server:~/Desktop/script$ nano service_control.sh
lens@lens-server:~/Desktop/script$ chmod +x service_control.sh
lens@lens-server:~/Desktop/script$ ./service_control.sh
=== Gestion des services ===
Entrer le nom du service (ex: ssh, apache2) : ssh
Action (start | stop | restart | status) : start
lens@lens-server:~/Desktop/script$ ./service_control.sh
=== Gestion des services ===
Entrer le nom du service (ex: ssh, apache2) : ssh
Action (start | stop | restart | status) : stop
Stopping 'ssh.service', but its triggering units are still active:
ssh.socket
lens@lens-server:~/Desktop/script$ ./service_control.sh
=== Gestion des services ===
Entrer le nom du service (ex: ssh, apache2) : ssh
Action (start | stop | restart | status) : status
O ssh.service - OpenBSD Secure Shell server
Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
Active: inactive (dead) since Fri 2026-02-06 18:41:37 UTC; 16s ago
Duration: 22min 657ms
TriggeredBy: ● ssh.socket
Docs: man:sshd(8)
      man:sshd_config(5)
      Process: 6076 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
      Process: 6077 ExecStart=/usr/sbin/sshd -D $SSH_OPTS (code=exited, status=0/SUCCESS)
      Main PID: 6077 (code=exited, status=0/SUCCESS)
      CPU: 91ms

Feb 06 18:19:36 lens-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 06 18:19:36 lens-server sshd[6077]: Server listening on 0.0.0.0 port 22.
Feb 06 18:19:36 lens-server sshd[6077]: Server listening on :: port 22.
Feb 06 18:19:36 lens-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 06 18:41:36 lens-server systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
```

4. Activer un service au démarrage

```
lens@lens-server:~/Desktop/script$ nano service_control.sh
lens@lens-server:~/Desktop/script$ sudo systemctl enable apache2
Synchronizing state of apache2.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable apache2
lens@lens-server:~/Desktop/script$
```

5. Afficher les logs pour un service et analyser

```
-- No entries --
lens@lens-server:~/Desktop/script$ sudo systemctl start ssh
lens@lens-server:~/Desktop/script$ sudo journalctl -u ssh
Feb 06 18:19:36 lens-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 06 18:19:36 lens-server sshd[6077]: Server listening on 0.0.0.0 port 22.
Feb 06 18:19:36 lens-server sshd[6077]: Server listening on :: port 22.
Feb 06 18:19:36 lens-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Feb 06 18:41:36 lens-server systemd[1]: Stopping ssh.service - OpenBSD Secure Shell server...
Feb 06 18:41:37 lens-server sshd[6077]: Received signal 15; terminating.
Feb 06 18:41:37 lens-server systemd[1]: ssh.service: Deactivated successfully.
Feb 06 18:41:37 lens-server systemd[1]: Stopped ssh.service - OpenBSD Secure Shell server.
Feb 06 19:02:26 lens-server systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Feb 06 19:02:27 lens-server sshd[7319]: Server listening on 0.0.0.0 port 22.
Feb 06 19:02:27 lens-server sshd[7319]: Server listening on :: port 22.
Feb 06 19:02:27 lens-server systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

Analyse

18:19:36: systemd démarre le service SSH, SSH est actif et prêt à recevoir des connexions.

18:41:36: system arrête le service, SSH a été désactivé volontairement

19:02:27: je relances SSH, SSH est de nouveau actif

6. Créer 2 utilisateurs et définir des mots de passe sécurisés

Création des utilisateurs

```
Feb 6 20:21
lens@lens-server: ~

lens@lens-server:~$ sudo adduser user1
[sudo] password for lens:
info: Adding user `user1' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user1' (1020) ...
info: Adding new user `user1' (1020) with group `user1 (1020)' ...
info: Creating home directory `/home/user1' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user1
Enter the new value, or press ENTER for the default
    Full Name []: jacque comper
    Room Number []: 5354646464
    Work Phone []: 34353435
    Home Phone []: 34538798
    Other []: ...
Is the information correct? [Y/n] y
info: Adding new user `user1' to supplemental / extra groups `users' ...
info: Adding user `user1' to group `users' ...
lens@lens-server:~$ sudo adduser user2
info: Adding user `user2' ...
info: Selecting UID/GID from range 1000 to 59999 ...
info: Adding new group `user2' (1021) ...
info: Adding new user `user2' (1021) with group `user2 (1021)' ...
info: Creating home directory `/home/user2' ...
info: Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for user2
Enter the new value, or press ENTER for the default
```

Vérification des utilisateurs

```
Feb 6 20:24
lens@lens-server: ~
lens@lens-server:~$ cut -d: -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
uucp
proxy
www-data
backup
list
irc
_apt
nobody
systemd-network
systemd-timesync
dhcpcd
messagebus
systemd-resolve
pollinate
polkitd
syslog
uidd
tcpdump
```

7. Vérifier et corriger les permissions des fichiers /etc/shadow et /etc/passwd

```
Feb 6 20:27
lens@lens-server: ~
mysql
user1
user2
lens@lens-server:~$ ls -l /etc/passwd /etc/shadow
-rw-r--r-- 1 root root 4011 Feb  6 20:13 /etc/passwd
-rw-r----- 1 root shadow 2316 Feb  6 20:13 /etc/shadow
lens@lens-server:~$ sudo chown root:root /etc/passwd /etc/shadow
lens@lens-server:~$ sudo chmod 644 /etc/passwd
lens@lens-server:~$ sudo chmod 640 /etc/shadow
```

Vérifier les fichiers avec permissions "world-writable"

```
lens@lens-server:~$ find / -type f -perm -o+w 2>/dev/null
/sys/kernel/security/apparmor/.remove
/sys/kernel/security/apparmor/.replace
/sys/kernel/security/apparmor/.load
/sys/kernel/security/apparmor/.notify
/sys/kernel/security/apparmor/.access
/proc/sys/kernel/ns_last_pid
/proc/pressure/io
/proc/pressure/cpu
/proc/pressure/memory
/proc/1/task/1/attr/current
/proc/1/task/1/attr/exec
/proc/1/task/1/attr/fscreate
/proc/1/task/1/attr/keycreate
/proc/1/task/1/attr/sockcreate
/proc/1/task/1/attr/smack/current
/proc/1/task/1/attr/apparmor/current
/proc/1/task/1/attr/apparmor/exec
/proc/1/attr/current
/proc/1/attr/exec
/proc/1/attr/fscreate
/proc/1/attr/keycreate
```

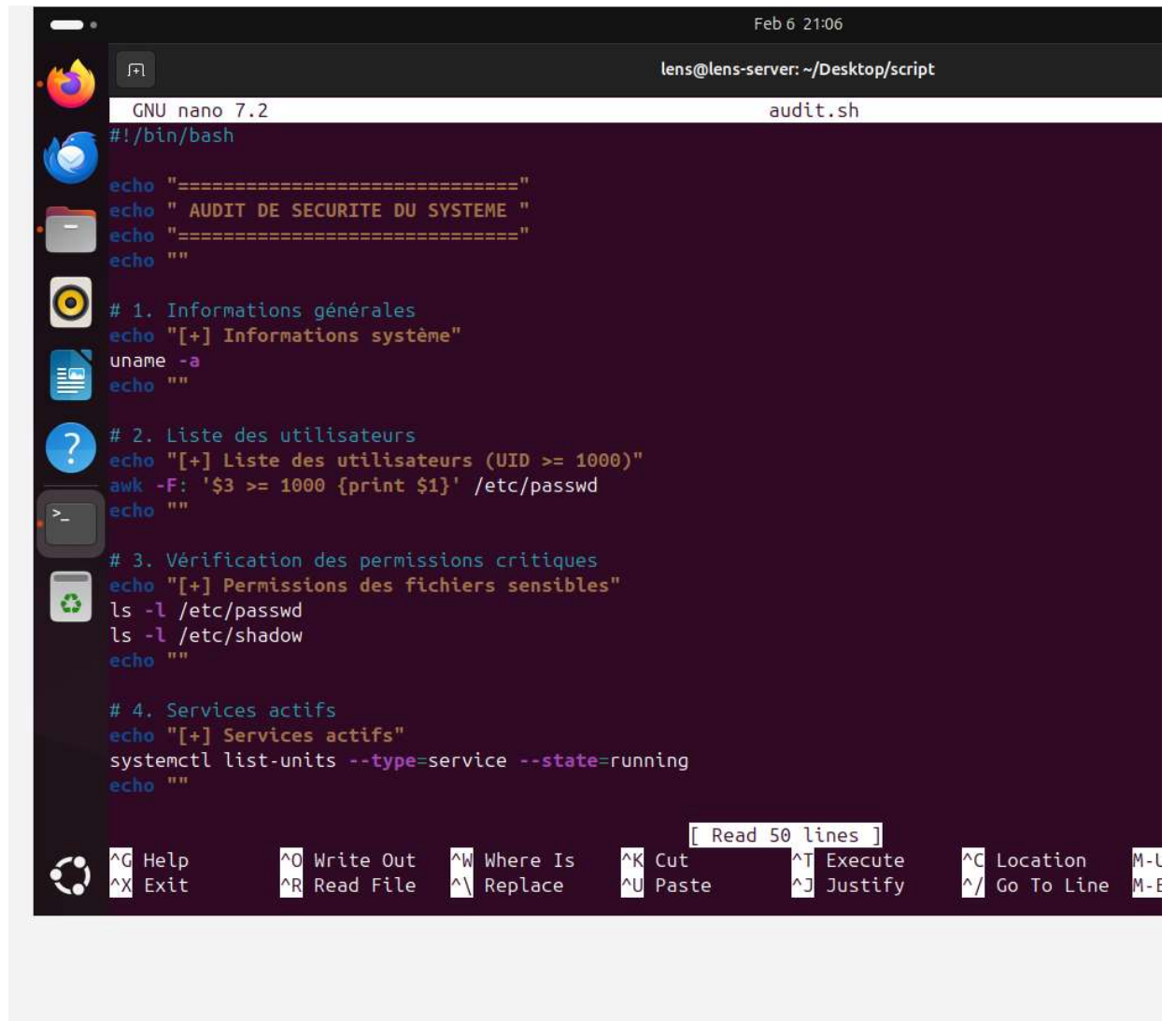
8. Activer et configurer le pare-feu UFW

```
lens@lens-server:~$ sudo apt install ufw
178 packages can be upgraded. Run 'apt list --upgradable' to see them.
lens@lens-server:~$ sudo apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.2-6).
0 upgraded, 0 newly installed, 0 to remove and 178 not upgraded.
lens@lens-server:~$ sudo ufw enable
Firewall is active and enabled on system startup
lens@lens-server:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
lens@lens-server:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
lens@lens-server:~$ sudo ufw status
Status: active

To Action From
--
Samba ALLOW Anywhere
Apache ALLOW Anywhere
Apache Full ALLOW Anywhere
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
Samba (v6) ALLOW Anywhere (v6)
Apache (v6) ALLOW Anywhere (v6)
Apache Full (v6) ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)
```

9. Auditer le système à l'aide du script audit.sh

Insertion de mon script dans nano avec le fichier audit.sh



```
Feb 6 21:06
lens@lens-server: ~/Desktop/script
GNU nano 7.2 audit.sh
#!/bin/bash

echo "===== "
echo " AUDIT DE SECURITE DU SYSTEME "
echo "===== "
echo ""

# 1. Informations générales
echo "[+] Informations système"
uname -a
echo ""

# 2. Liste des utilisateurs
echo "[+] Liste des utilisateurs (UID >= 1000)"
awk -F: '$3 >= 1000 {print $1}' /etc/passwd
echo ""

# 3. Vérification des permissions critiques
echo "[+] Permissions des fichiers sensibles"
ls -l /etc/passwd
ls -l /etc/shadow
echo ""

# 4. Services actifs
echo "[+] Services actifs"
systemctl list-units --type=service --state=running
echo ""

[ Read 50 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   M-L
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E
```

Execution de mon script pour l'audit

```
lens@lens-server: ~/Desktop/script
lens@lens-server:~/Desktop/script$ nano audit.sh
lens@lens-server:~/Desktop/script$ chmod +x audit.sh
lens@lens-server:~/Desktop/script$ ./audit.sh
=====
AUDIT DE SECURITE DU SYSTEME
=====

[+] Informations système
Linux lens-server 6.8.0-88-generic #89-Ubuntu SMP PREEMPT_DYNAMIC Sat Oct 11 01:02:46 UTC 2025 x86_64 x86_64 x86_64 GNU
Linux

[+] Liste des utilisateurs (UID >= 1000)
nobody
lens
Tom
jean
jaque
marie
martine
flore
joseph
ralph
junior
josue
paul
bruno
andre
islande
carla
admin_temp
user1
user2
```

```
lens@lens-server: ~/Desktop/script

[+] Permissions des fichiers sensibles
-rw-r--r-- 1 root root 4011 Feb  6 20:13 /etc/passwd
-rw-r----- 1 root root 2316 Feb  6 20:13 /etc/shadow

[+] Services actifs
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
apache2.service	loaded	active	running	The Apache HTTP Server
auditd.service	loaded	active	running	Security Auditing Service
avahi-daemon.service	loaded	active	running	Avahi mDNS/DNS-SD Stack
colord.service	loaded	active	running	Manage, Install and Generate Color Profiles
cron.service	loaded	active	running	Regular background program processing daemon
cups-browsed.service	loaded	active	running	Make remote CUPS printers available locally
cups.service	loaded	active	running	CUPS Scheduler
dbus.service	loaded	active	running	D-Bus System Message Bus
fwupd.service	loaded	active	running	Firmware update daemon
gdm.service	loaded	active	running	GNOME Display Manager
gnome-remote-desktop.service	loaded	active	running	GNOME Remote Desktop
kerneloops.service	loaded	active	running	Tool to automatically collect and submit kernel crash signatures
ModemManager.service	loaded	active	running	Modem Manager
multipathd.service	loaded	active	running	Device-Mapper Multipath Device Controller
mysql.service	loaded	active	running	MySQL Community Server
named.service	loaded	active	running	BIND Domain Name Server
NetworkManager.service	loaded	active	running	Network Manager
nnbd.service	loaded	active	running	Samba NMB Daemon
polkit.service	loaded	active	running	Authorization Manager
power-profiles-daemon.service	loaded	active	running	Power Profiles daemon
rsyslog.service	loaded	active	running	System Logging Service
rtkit-daemon.service	loaded	active	running	RealtimeKit Scheduling Policy Service
snbd.service	loaded	active	running	Samba SMB Daemon
snapd.service	loaded	active	running	Snap Daemon
switcheroo-control.service	loaded	active	running	Switcheroo Control Proxy service
systemd-journald.service	loaded	active	running	Journal Service

10. Lister les 10 processus les plus gourmands en CPU et RAM

En CPU

```
lens@lens-server:~/Desktop/script$ ps aux --sort=-%cpu | head -n 11
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
lens          6270 42.8  0.2  11148 4736 pts/0    R+   21:17   0:00 ps aux --sort=-%cpu
lens          2056  4.0 10.5 3545868 212496 ?        Ssl  19:58   3:14 /usr/bin/gnome-shell
lens          2693  2.3 16.4 11869152 332212 ?        Sl   19:58   1:51 /snap/firefox/7766/usr/lib/firefox/firefox
lens          3284  2.1 14.2 3156324 288184 ?        Sl   19:58   1:43 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:42333 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {cae26ec5-aae6-4ca9-806b-5c79c23020ea} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
fox/browser/omni.ja -appDir /snap/firefox/7766/usr/lib/firefox/browser 6 tab
mysql          1203  1.1  0.3 1327756 7408 ?        Ssl  19:56   0:58 /usr/sbin/mysqld
lens          3944  0.3  2.5 556244 50824 ?        Ssl  20:11   0:13 /usr/libexec/gnome-terminal-server
root          4209  0.3  0.0 0 0 ?        I    20:28   0:09 [kworker/u2:2-events_unbound]
lens          6025  0.2 12.0 1130740 243804 ?        Sl   20:59   0:03 /usr/bin/nautilus --gapplication-service
root          6172  0.2  0.0 0 0 ?        I    21:06   0:01 [kworker/u2:4-events_unbound]
root          4881  0.2  0.0 0 0 ?        I    20:41   0:04 [kworker/u2:0-events_power_efficient]

lens@lens-server:~/Desktop/script$ ps aux --sort=-%mem | head -n 11
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
lens          2693  2.3 16.4 11869184 332248 ?        Sl   19:58   1:52 /snap/firefox/7766/usr/lib/firefox/firefox
lens          3284  2.1 14.4 3156324 291760 ?        Sl   19:58   1:44 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:42333 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {cae26ec5-aae6-4ca9-806b-5c79c23020ea} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
```

En RAM

```
Feb 6 21:20
lens@lens-server:~/Desktop/script$ ps aux --sort=-%mem | head -n 11
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root          6172  0.2  0.0 0 0 ?        I    21:06   0:01 [kworker/u2:4-events_unbound]
root          4881  0.2  0.0 0 0 ?        I    20:41   0:04 [kworker/u2:0-events_power_efficient]

lens@lens-server:~/Desktop/script$ ps aux --sort=-%mem | head -n 11
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
lens          2693  2.3 16.4 11869184 332248 ?        Sl   19:58   1:52 /snap/firefox/7766/usr/lib/firefox/firefox
lens          3284  2.1 14.4 3156324 291760 ?        Sl   19:58   1:44 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:42333 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {cae26ec5-aae6-4ca9-806b-5c79c23020ea} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
fox/browser/omni.ja -appDir /snap/firefox/7766/usr/lib/firefox/browser 6 tab
lens          6025  0.2 12.0 1130740 243804 ?        Sl   20:59   0:03 /usr/bin/nautilus --gapplication-service
lens          2056  4.1 10.5 3545868 212496 ?        Ssl  19:58   3:15 /usr/bin/gnome-shell
lens          2875  0.1  3.8 2635892 78344 ?        Sl   19:58   0:04 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:35775 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {8dad8946-3ca7-4283-968d-73e1f79a22b2} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
fox/browser/omni.ja -appDir /snap/firefox/7766/usr/lib/firefox/browser 3 tab
lens          3944  0.3  2.5 556244 50824 ?        Ssl  20:11   0:13 /usr/libexec/gnome-terminal-server
lens          3095  0.0  2.2 2606004 46276 ?        Sl   19:58   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:35775 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {c1f532c8-ed0c-4c33-ac3e-fc957f2e2b0e} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
fox/browser/omni.ja -appDir /snap/firefox/7766/usr/lib/firefox/browser 5 tab
lens          2634  0.0  1.7 2803220 35248 ?        Sl   19:58   0:01 gjs /usr/share/gnome-shell/extensions/ding@rastersoft
.com/app/ding.js -E -P /usr/share/gnome-shell/extensions/ding@rastersoft.com/app
lens          3479  0.0  1.7 2560308 34880 ?        Sl   19:58   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:43206 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {bb668945-9b93-46ee-9804-f4811836cf55} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
fox/browser/omni.ja -appDir /snap/firefox/7766/usr/lib/firefox/browser 9 tab
lens          3470  0.0  1.7 2560308 34844 ?        Sl   19:58   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:43206 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {c539017d-3047-43dh-9766-5d2e53f29b98} -parentPid 2693 -cra
```

11. Vérifier l'espace disque et détecter les partitions saturées

```
141M    /var/log
lens@lens-server:~/Desktop/script$ free -h
              total        used        free      shared  buff/cache   available
Mem:           1.9Gi        1.2Gi          99Mi        24Mi       797Mi       745Mi
Swap:          2.0Gi        904Mi        1.1Gi

lens@lens-server:~/Desktop/script$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            197M  4.6M  193M   3% /run
/dev/mapper/ubuntu--vg-ubuntu--lv 12G   11G  226M  98% /
tmpfs            985M    0  985M   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
/dev/sda2        2.0G  109M   1.7G   6% /boot
tmpfs            197M  124K  197M   1% /run/user/1000

lens@lens-server:~/Desktop/script$
```

12. Lister tous les ports ouverts et services réseau actifs

Ports actifs

```
lens@lens-server:~/Desktop/script$ ss -tulnp
Netid  State  Recv-Q  Send-Q      Local Address:Port      Peer Address:Port      Process
udp    UNCONN 0        0      192.168.56.105:53        0.0.0.0:*               firewalld
udp    UNCONN 0        0      10.0.2.15:53            0.0.0.0:*               firewalld
udp    UNCONN 0        0      127.0.0.1:53            0.0.0.0:*               firewalld
udp    UNCONN 0        0      127.0.0.54:53           0.0.0.0:*               firewalld
udp    UNCONN 0        0      127.0.0.53%lo:53        0.0.0.0:*               firewalld
udp    UNCONN 0        0      192.168.56.255:137      0.0.0.0:*               firewalld
udp    UNCONN 0        0      192.168.56.105:137      0.0.0.0:*               firewalld
udp    UNCONN 0        0      10.0.2.255:137          0.0.0.0:*               firewalld
udp    UNCONN 0        0      10.0.2.15:137           0.0.0.0:*               firewalld
udp    UNCONN 0        0      0.0.0.0:137             0.0.0.0:*               firewalld
udp    UNCONN 0        0      192.168.56.255:138      0.0.0.0:*               firewalld
udp    UNCONN 0        0      192.168.56.105:138      0.0.0.0:*               firewalld
udp    UNCONN 0        0      10.0.2.255:138          0.0.0.0:*               firewalld
udp    UNCONN 0        0      10.0.2.15:138           0.0.0.0:*               firewalld
udp    UNCONN 0        0      0.0.0.0:138             0.0.0.0:*               firewalld
udp    UNCONN 0        0      0.0.0.0:5353            0.0.0.0:*               firewalld
udp    UNCONN 0        0      0.0.0.0:42973           0.0.0.0:*               firewalld
udp    UNCONN 0        0      [::]:53269              [::]:*                  firewalld
udp    UNCONN 0        0      [::1]:53                [::]:*                  firewalld
udp    UNCONN 0        0      [fe80::a00:27ff:fe6b:6cc5]%enp0s3:53 [::]:*                  firewalld
udp    UNCONN 0        0      [fe80::5b14:5d4:dd4a:f49f]%enp0s8:53 [::]:*                  firewalld
udp    UNCONN 0        0      [fd17:625c:f037:2:c942:6167:71ad:22a]:53 [::]:*                  firewalld
udp    UNCONN 0        0      [fd17:625c:f037:2:a00:27ff:fe6b:6cc5]:53 [::]:*                  firewalld
udp    UNCONN 0        0      [::]:5353                [::]:*                  firewalld
tcp    LISTEN 0        4096      127.0.0.54:53            0.0.0.0:*               firewalld
tcp    LISTEN 0        10      127.0.0.1:53            0.0.0.0:*               firewalld
tcp    LISTEN 0        10      10.0.2.15:53            0.0.0.0:*               firewalld
tcp    LISTEN 0        151      127.0.0.1:3306           0.0.0.0:*               firewalld
tcp    LISTEN 0        4096      127.0.0.53%lo:53        0.0.0.0:*               firewalld
tcp    LISTEN 0        10      192.168.56.105:53        0.0.0.0:*               firewalld
```

Réseaux actifs

```
Feb 6 21:28
lens@lens-server: ~/Desktop/script
no vm guests are running outdated hypervisor (qemu) binaries on this host.
lens@lens-server:~/Desktop/script$ netstat -tulnp
(Not all processes could be identified, non-owned process info
will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.54:53          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:53          0.0.0.0:*               LISTEN      -
tcp        0      0 10.0.2.15:53          0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:3306        0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.53:53         0.0.0.0:*               LISTEN      -
tcp        0      0 192.168.56.105:53     0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:33060       0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:631         0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:139           0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:22            0.0.0.0:*               LISTEN      -
tcp        0      0 0.0.0.0:445           0.0.0.0:*               LISTEN      -
tcp        0      0 127.0.0.1:953         0.0.0.0:*               LISTEN      -
tcp6       0      0 fe80::a00:27ff:fe6b::53 :::*                   LISTEN      -
tcp6       0      0 :::1:953              :::*                   LISTEN      -
tcp6       0      0 fd17:625c:f037:2:c94:53 :::*                   LISTEN      -
tcp6       0      0 :::1:631              :::*                   LISTEN      -
tcp6       0      0 :::139                :::*                   LISTEN      -
tcp6       0      0 :::80                 :::*                   LISTEN      -
tcp6       0      0 :::22                 :::*                   LISTEN      -
tcp6       0      0 fe80::5b14:5d4:dd4a::53 :::*                   LISTEN      -
tcp6       0      0 fd17:625c:f037:2:a00:53 :::*                   LISTEN      -
tcp6       0      0 :::445                :::*                   LISTEN      -
tcp6       0      0 :::1:53               :::*                   LISTEN      -
udp        0      0 192.168.56.105:53     0.0.0.0:*               -           -
udp        0      0 10.0.2.15:53          0.0.0.0:*               -           -
udp        0      0 127.0.0.1:53          0.0.0.0:*               -           -
udp        0      0 127.0.0.54:53         0.0.0.0:*               -           -
```

13. Écrire un script Bash pour collecter les performances toutes les 5 min

Creation dans nano et execution de mon script

```

lens@lens-server: ~/Desktop/script$ nano /usr/local/bin/perf_collect.sh
lens@lens-server: ~/Desktop/script$ sudo chmod +x /usr/local/bin/perf_collect.sh
lens@lens-server: ~/Desktop/script$ sudo /usr/local/bin/perf_collect.sh
[CPU]
[MEMOIRE]
[DISQUE]
[TOP CPU]
[TOP RAM]
lens@lens-server: ~/Desktop/script$ sudo tail /var/log/performance.log
lens      3284   3.0  16.3 3339336 330484 ?      SL   19:58   3:22 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:42333 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {cae26ec5-aae6-4ca9-806b-5c79c23020ea} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
fox/browser/omni.ja -appDir /snap/firefox/7766/usr/lib/firefox/browser 6 tab
lens      2693   2.6  15.2 11891900 306824 ?      SL   19:58   3:00 /snap/firefox/7766/usr/lib/firefox/firefox

USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
lens      3284   3.0  16.3 3339336 330484 ?      SL   19:58   3:22 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:42333 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {cae26ec5-aae6-4ca9-806b-5c79c23020ea} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
fox/browser/omni.ja -appDir /snap/firefox/7766/usr/lib/firefox/browser 6 tab
lens      2693   2.6  15.2 11891900 306824 ?      SL   19:58   3:00 /snap/firefox/7766/usr/lib/firefox/firefox
lens      6025   0.1  12.0 1130740 243292 ?      SL   20:59   0:03 /usr/bin/nautilus --gapplication-service
lens      2056   4.8   9.5 3545868 191808 ?      Rsl  19:58   5:26 /usr/bin/gnome-shell
lens      2875   0.1   3.9 2646116 80548 ?      SL   19:58   0:07 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
oc -isForBrowser -prefsHandle 0:35775 -prefMapHandle 1:277097 -jsInitHandle 2:223968 -parentBuildID 20260204004611 -sand
boxReporter 3 -chrootClient 4 -ipcHandle 5 -initialChannelId {8dad8946-3ca7-4283-968d-73e1f79a22b2} -parentPid 2693 -cra
shReporter 6 -crashHelper 7 -greomni /snap/firefox/7766/usr/lib/firefox/omni.ja -appomni /snap/firefox/7766/usr/lib/fire
fox/browser/omni.ja -appDir /snap/firefox/7766/usr/lib/firefox/browser 3 tab

```

```

Feb 6 22:00
lens@lens-server: ~/Desktop/script

GNU nano 7.2 /usr/local/bin/perf_collect.sh
#!/bin/bash

LOGFILE="/var/log/performance.log"

echo "===== " >> "$LOGFILE"
echo "Collecte des performances - $(date)" >> "$LOGFILE"
echo "===== " >> "$LOGFILE"

echo "[CPU]"
uptime >> "$LOGFILE"
echo "" >> "$LOGFILE"

echo "[MEMOIRE]"
free -h >> "$LOGFILE"
echo "" >> "$LOGFILE"

echo "[DISQUE]"
df -h >> "$LOGFILE"
echo "" >> "$LOGFILE"

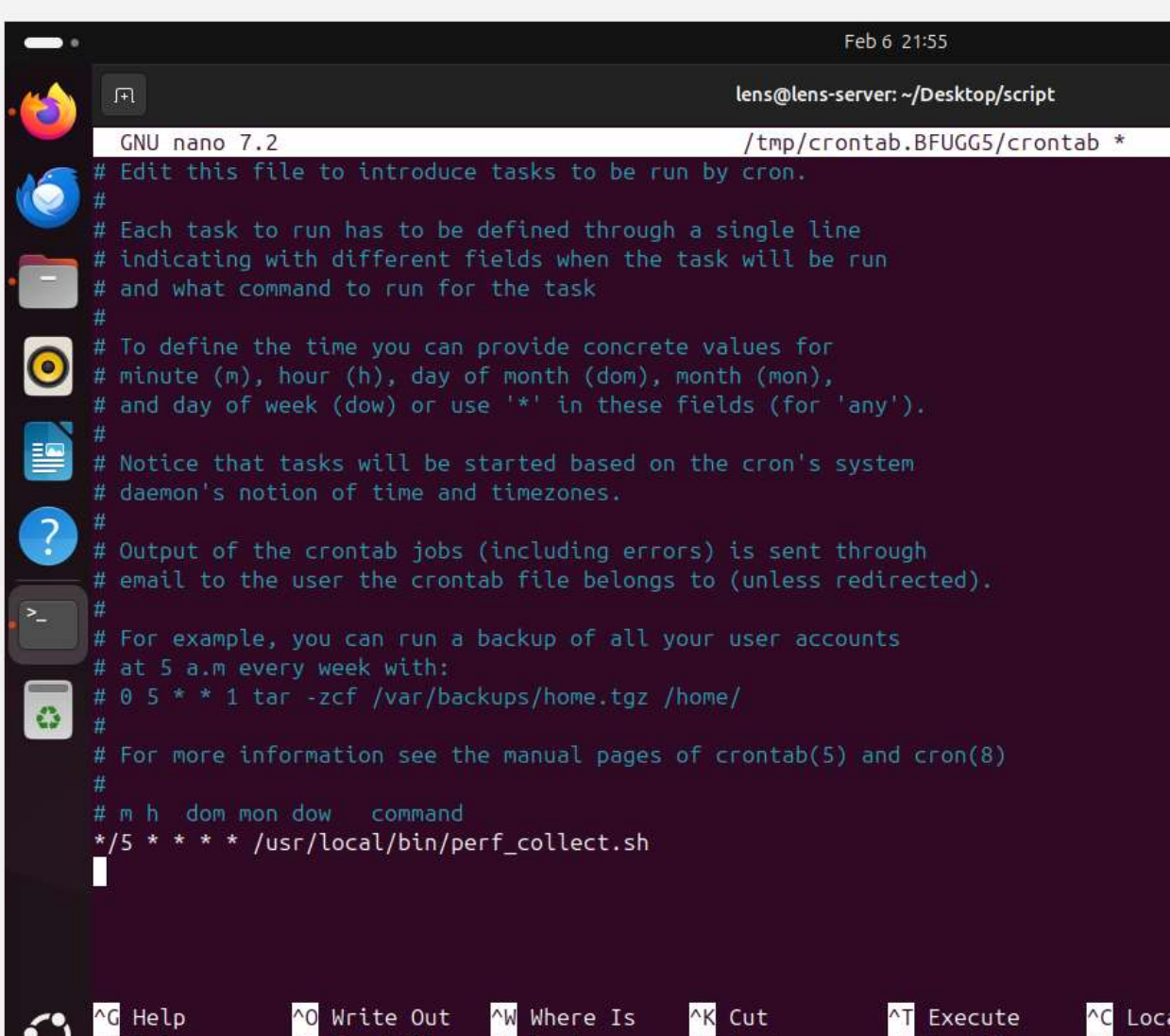
echo "[TOP CPU]"
ps aux --sort=-%cpu | head -n 6 >> "$LOGFILE"
echo "" >> "$LOGFILE"

echo "[TOP RAM]"
ps aux --sort=-%mem | head -n 6 >> "$LOGFILE"
echo "" >> "$LOGFILE"

[ Read 27 lines ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/

```

Ajouter une tâche a cron et verification que la tache est enrregistré



```
Feb 6 21:55
lens@lens-server: ~/Desktop/script
GNU nano 7.2 /tmp/crontab.BFUGG5/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow   command
*/5 * * * * /usr/local/bin/perf_collect.sh
```

```
bash: /5: No such file or directory
lens@lens-server:~/Desktop/script$ sudo crontab -e
no crontab for root - using an empty one
crontab: installing new crontab
lens@lens-server:~/Desktop/script$ sudo crontab -l
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow  command
*/5 * * * * /usr/local/bin/perf_collect.sh
lens@lens-server:~/Desktop/script$ sudo nano /usr/local/bin/perf_collect.sh
lens@lens-server:~/Desktop/script$
```

Donc oui le script est exécuté tous le cinq minute automatiquement

Cocclusion.

Ce travail m'a permis de comprendre ce qu'est un service / démon sous Linux, savoir lister, démarrer, arrêter, activer et désactiver des services, savoir lister, démarrer, arrêter, activer et désactiver des services, automatiser la gestion des services via scripts, Mettre en place des comptes sécurisés et permissions correctes configurer le pare-feu et services essentiels analyser les logs et détecter les anomalies comprendre les ressources critiques du système : CPU, RAM, disque, réseau utiliser les outils Linux pour mesurer la charge système automatiser la collecte et la surveillance avec scripts Bash. Donc la tâche est réussie.